

Lesson Plan

- 1) Finish linear independence
- 2) Subspace
- 3) Eigenvalue decomposition

def Given $x_1, \dots, x_k$, they are linearly independent if
 $\Rightarrow c_1, \dots, c_k$ that $\underbrace{\sum_{i=1}^k c_i x_i = 0}_{CTX \text{ or } \langle C, X \rangle = 0}$ implies $c_1 = \dots = c_k = 0$.

Example. $x_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 4 \end{bmatrix}$, $x_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}$, $x_3 = \begin{bmatrix} -1 \\ -2 \\ 1 \\ 1 \end{bmatrix}$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 2 & 1 & -2 & 0 \\ -3 & 0 & 1 & 0 \\ 4 & 2 & 1 & 0 \end{array} \right] \xrightarrow[-4R_1]{-2R_1, +3R_1, -4R_1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -1 & 0 & -2 \\ 0 & -3 & -2 & -3 \\ 0 & -2 & 5 & -4 \end{array} \right] \xrightarrow[-2R_2]{x+1, +3R_2, +2R_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 5 & -8 \end{array} \right] \xrightarrow[-5R_3]{x \cdot \frac{1}{2}} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$c_1 x_1 + c_2 x_2 + c_3 x_3 = 0$$

$$\Rightarrow c_1 = c_2 = c_3 = 0$$

$\therefore$ linearly independent.

Example $M_1, \dots, M_k \in \mathbb{R}^{n \times n}$ are linearly independent.

$$c_1 M_1 + \dots + c_k M_k = 0 \quad \text{iff} \quad c_1 = \dots = c_k = 0.$$

Given M_1, M_2 they are linearly dependent if $M_1 = c M_2$ for some $c \neq 0$.

Vector space

$V = (\underbrace{\mathcal{S}}_{\text{set}}, +, \cdot)$ if it satisfies conditions 1-10

$$+ : V \times V \rightarrow V$$

$$\cdot : \mathbb{R} \times V \rightarrow V$$

$\hookrightarrow$ Field " $\mathbb{R}$ ": $F \times V \rightarrow V$

Subspace

Given vector space $V = (V, +, \cdot)$

Then $U \subseteq V$ is a subspace with $+$, $\cdot$ st. $U \neq \emptyset$ and $(U, +, \cdot)$ is a vector space

U is a proper subspace of V if $U \neq V$.

$$U \subset V$$

Example

- 1) $U = V$
- 2) $U = \{0\}$ trivial subspace
- 3) Given linear transformation

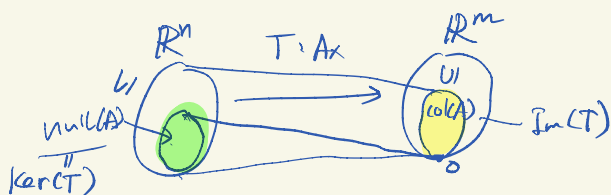
$$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$T(x) = Ax \quad A \in \mathbb{R}^{m \times n}$$

$$A = [\underbrace{a_1 \dots a_n}]^m \quad a_i \in \mathbb{R}^m$$

$$\text{Then } \underline{\text{col}(A)} = \left\{ \underbrace{c_1 a_1 + \dots + c_n a_n}_{\in \mathbb{R}^m} \mid c_1, \dots, c_n \in \mathbb{R} \right\} \subseteq \mathbb{R}^m$$

$$\text{null}(A) = \{ x \in \mathbb{R}^n \mid Ax = 0 \} \subseteq \mathbb{R}^n$$



For space V , 1) $S = \{v_1, \dots, v_n\} \subset V$ is a spanning set of V if $\text{Span}(S) = V$

2) S is a minimal spanning set of V if V can be generated by any $S' \subset S$ ($S' \neq S$)

Example $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -1 \end{bmatrix} \quad V = \text{col}(A)$
 $= [v_1 \ v_2 \ v_3] = v_1 - v_2$

$$\text{Span}(\{v_1, v_2\}) = \text{col}(A) = \text{Span}(\{v_1, v_2, v_3\})$$

↑ a minimal spanning set

$\{v_1, v_2\}$ is a basis of V .

Basis

$B = \{v_1, \dots, v_n\}$ is a basis of V if

- 1) $v_1, \dots, v_n$ are linearly independent
- 2) $v_1, \dots, v_n$ span V

Dim . V has a basis of n vectors $\dim(V) = n$.

* $\{0\}$ has dimension 0.

Rank-Nullity Theorem

For any linear $T: V \rightarrow W$ (U is finite)

$$\dim(U) = \underbrace{\text{rank}(T)}_{\dim(\text{Im}(T))} + \underbrace{\text{nullity}(T)}_{\dim(\text{Ker}(T))}$$

Suppose $T(x) = Ax$, $A \in \mathbb{R}^{n \times m}$

$$n = \underline{\underline{\dim(U)}} = \text{rank}(A) + \text{nullity}(A)$$

Eigenvalue Decomposition

$A \in \mathbb{R}^{n \times n}$ symmetric $A = U \Lambda U^T$ where $U^T U = I$, Λ is diagonal

Rayleigh-Ritz Theorem

① Given symmetric $A \in \mathbb{R}^{n \times n}$, solution to the problem

$$\min_{x \in \mathbb{R}^n} x^T A x$$

$$\text{s.t. } \|x\| = 1$$

is the smallest eigenvector of A .

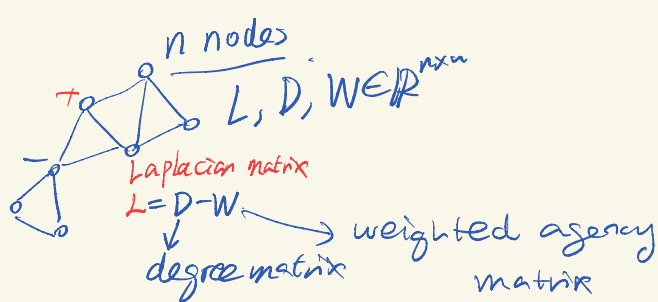
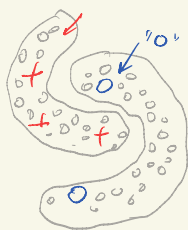
application spectral clustering

② $X = [x_1, \dots, x_k]$, $x_i \in \mathbb{R}^n$ ($k \leq n$), solution to minimization problem are k smallest eigenvector of A .

$$\min_{X \in \mathbb{R}^{n \times k}} \text{tr}(X^T A X)$$

$$\text{s.t. } X^T X = I_k \quad (k \times k)$$

Ex. half-moon dataset



$$LCE + \lambda \frac{\text{tr}(X^T L X)}{\text{manifold regularization}}$$